

3VD51203-3D6A-7H, HongZhanpeng, 20250610

REV.	ECN. NO.	APPD.
A	BC-14-0065178	Adward Zheng
B	BC-16-0017921	LIANG GONG
BX1	1/17'18	Davy Bu
BX2	1/19'18	Davy Bu

PRODUCT NO.: 3V * * * * *

DISPLAY PORT SERIES

PRODUCT SERIES:
D: STANDARD TYPE

TERMINATION TYPE:
5: R/A SMT

CONNECTOR TYPE:
1: RECEPTACLE
NO. OF CONTACT POS.:
20: 20 POS

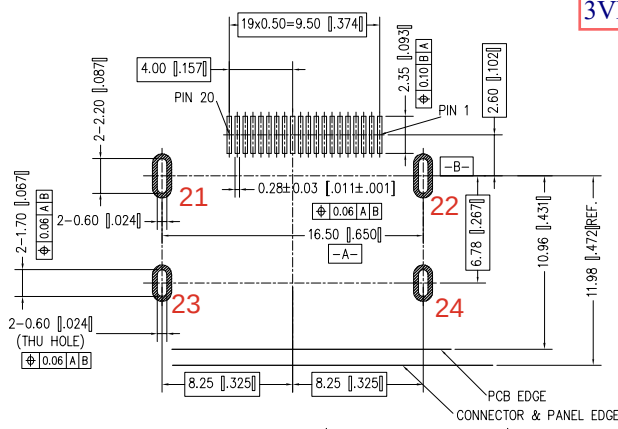
HALOGEN-FREE CODE

PACKAGE CODE:
4: SOFT TRAY
7: TAPE REEL

EXTENSION CODE

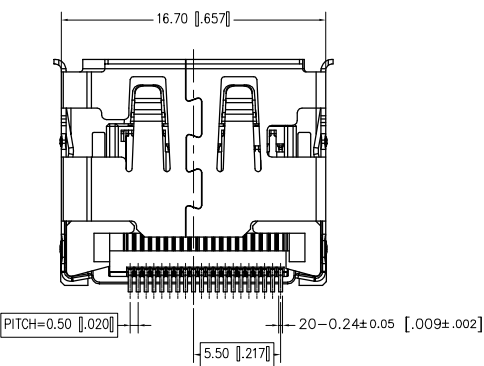
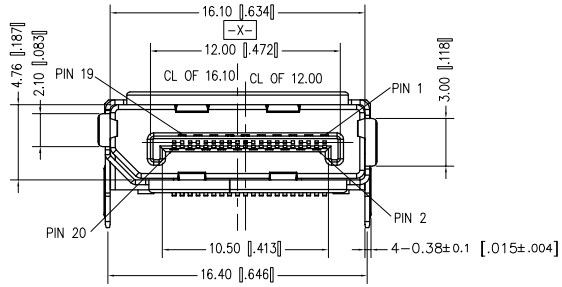
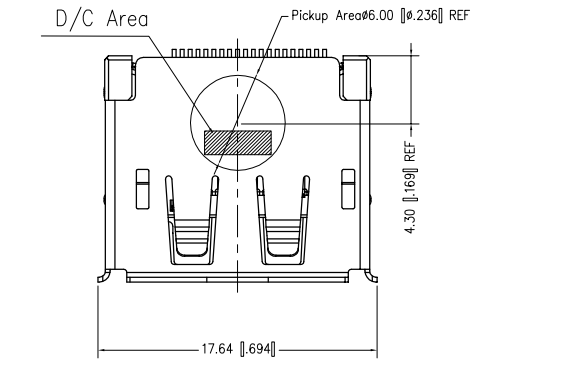
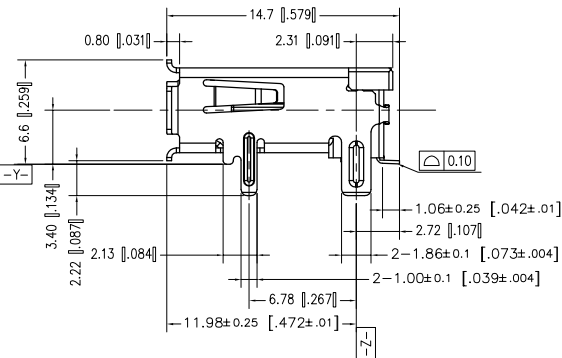
CONTACT GOLD PLATING:
1: 1u" GOLD
3: 30u" GOLD
7: 15u" GOLD

- NOTES:
- ELECTRICAL CHARACTERISTICS:
1-1. CONTACT CURRENT RATING: 0.5A MIN.
FINGER CURRENT RATING: 1.0A MIN.
APPLIED VOLTAGE RATING: 40V AC
1-2. CONTACT RESISTANCE:
CONTACT RESISTANCE: 30 MILLIOHMS MAXIMUM.
AFTER TEST: CHANGE FROM INITIAL VALUE 30MILLIOHMS MAX.
1-3. SHELL RESISTANCE:
AFTER TEST: CHANGE FROM INITIAL VALUE 50MILLIOHMS MAX.
1-4. DIELECTRIC WITHSTANDING VOLTAGE: 500V AC RMS.
1-5. INSULATION RESISTANCE:
UNMATED PLUG: 100 MEGAOHMS MIN. AT 500V DC.
MATED PLUG: 10 MEGAOHMS MIN. AT 150V DC.
 - MECHANICAL CHARACTERISTICS:
2-1. CONNECTOR MATING FORCE: 4.5Kgf MAX.
2-2. CONNECTOR UNMATING FORCE: 1.0~4.0 Kgf.
2-3. DURABILITY: 10000 CYCLES.
2-4. OPERATING TEMPERATURE RANGE: -40~85°C;
HUMIDITY: 85% RH MAX.
 - RECOMMENDED PROCESS:
3-1. IR REFLOW, PEAK TEMPERATURE 250°C, 10SEC MAX.
3-2. COPLANARITY OF SMT TERMINAL IS 0.10mm MAX.
 - RoHS AND HALOGEN-FREE COMPLIANT.



RECOMMENDED P.C.B. LAYOUT (COMPONENT SIDE VIEW)

PCB THICKNESS: 1.60±0.05mm
DEFAULT TOLERANCE: ±0.05



PIN NUMBER	SOURCE-SIDE PIN ASSIGNMENT	SINK-SIDE PIN ASSIGNMENT
1	ML_Lane 0(p)	ML_Lane 3(n)
2	GND	GND
3	ML_Lane 0(n)	ML_Lane 3(p)
4	ML_Lane 1(p)	ML_Lane 2(n)
5	GND	GND
6	ML_Lane 1(n)	ML_Lane 2(p)
7	ML_Lane 2(p)	ML_Lane 1(n)
8	GND	GND
9	ML_Lane 2(n)	ML_Lane 1(p)
10	ML_Lane 3(p)	ML_Lane 0(n)
11	GND	GND
12	ML_Lane 3(n)	ML_Lane 0(p)
13	CONFIG1	CONFIG1
14	CONFIG2	CONFIG2
15	AUX CH (p)	AUX CH (p)
16	GND	GND
17	AUX CH (n)	AUX CH (n)
18	HOT PLUG DETECT	HOT PLUG DETECT
19	RETURN	RETURN
20	DP PWR	DP PWR

③	CONTACT	20	COPPER ALLOY	NIKEL UNDERPLATING OVERALL, 30u" MIN. AU PLATING AT CONTACT AREA, TIN ALLOY(LEAD FREE) PLATING AT SOLDER AREA.
②	HOUSING	1	HIGH TEMPERATURE THERMOPLASTIC UL94V-0	COLOR: BLACK
①	SHELL	1	STAINLESS STEEL	NIKEL UNDERPLATING OVERALL, TIN ALLOY(LEAD FREE) PLATING AT SOLDER AREA.
ITEM	DESCRIPTION	Q'TY	MATERIAL	REMARK
X.±	X.±	UNITS	mm INCH	NAME(INTENDED USE)
.X± 0.30 .015	.X'±	MAT'L	N/A	CUSTOMER
.XX± 0.25 .010	.XX'±	FINISH	N/A	PART NO.(INTENDED USE)
.XXX±	.XXX'±	Q'TY	N/A	3VD5120*-3D6A-7H
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				CHKD: Z.J.Chen 1/19'18
				DRAW: Noah Ji 1/19'18
				DWG NO.: 301-0000-3122
				SCALE: N/A
				SHEET: 1/1
				REV. BX2



CLASS: ☐CONFIDENTIAL ☐SECRET ☒GENERAL

TITLE: DISPLAYPORT CONNECTOR SMT TYPE

